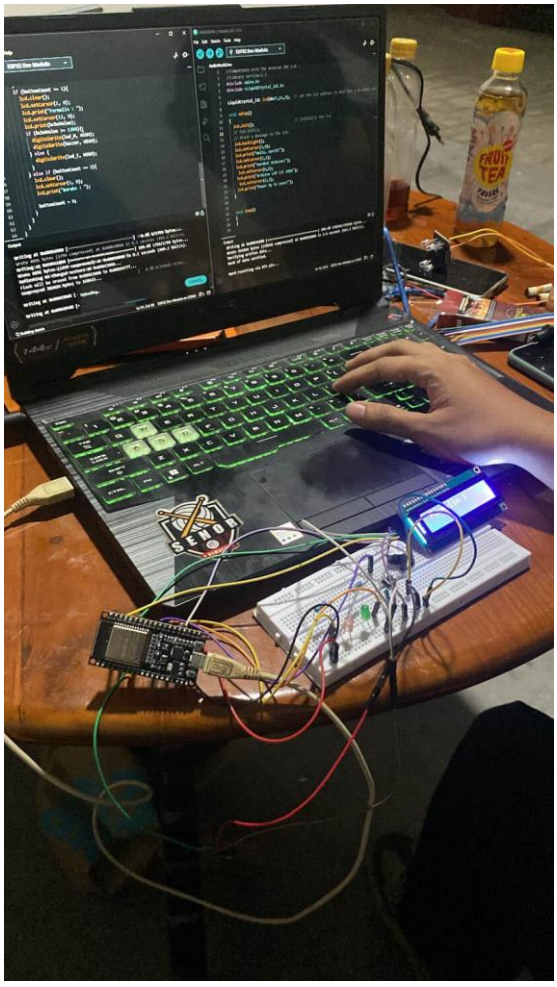
PORTOFOLIO PROJECT

Name : Putra Erik Ibrahim

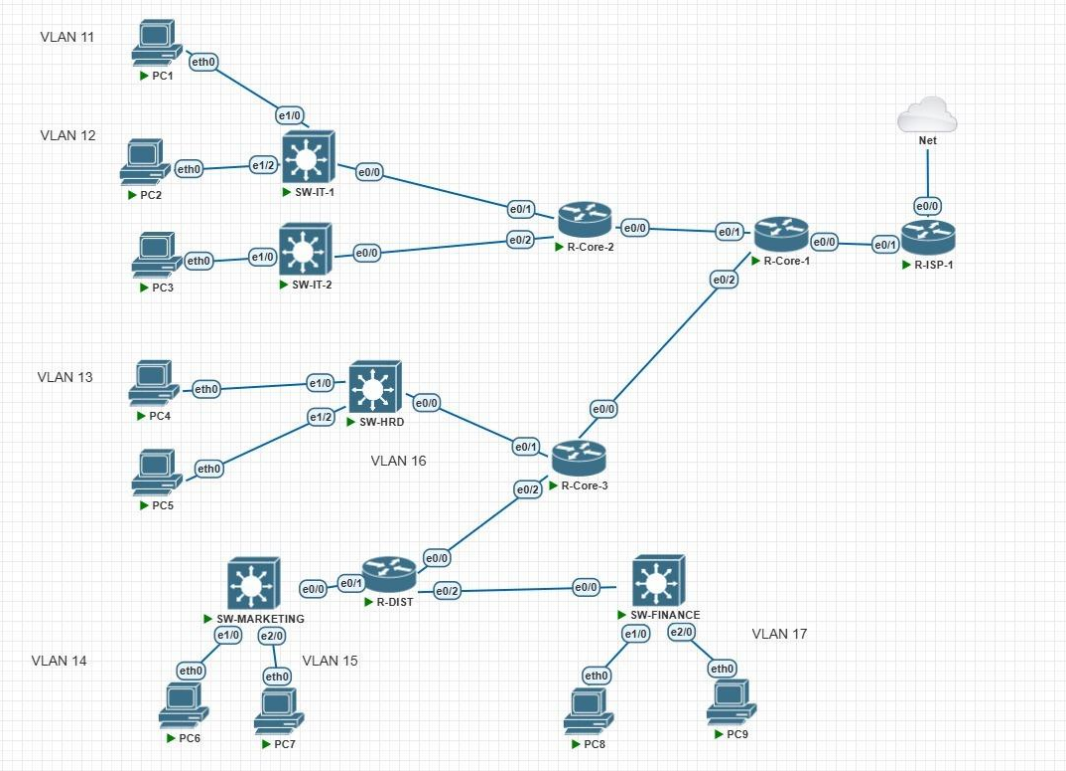
NIM : 241344058

1. **Alat pendeteksi formalin dan boraks embedded system menggunakan ESP32**





2. Routing OSPF



a.

Perangkat	Fungsi
R-Core-1	Berfungsi sebagai router inti pada OSPF Area 0 yang menghubungkan R-Core-2 dan R-Core-3. R-Core-1 juga menjadi penghubung antara jaringan internal OSPF dengan R-ISP-1 untuk akses menuju jaringan eksternal/Int
R-Core-2	Berfungsi sebagai router penghubung jaringan pada sisi Area 1 dengan backbone Area 0. R-Core-2 menangani komunikasi menuju VLAN 11 dan VLAN 12 melalui SW-IT-1 dan SW-IT-2 serta meneruskan informasi routing ke R-Core-1.
R-Core-3	Berfungsi sebagai router penghubung antara Area 0, Area 1, Area 2. R-Core 3 menghubungkan jaringan VLAN 13 melalui SW-HRD serta menghubungkan jaringan Area 2 melalui R-DIST.
R-DIST	Berfungsi sebagai router distribusi pada OSPF Area 2 yang menghubungkan R-Core-3 dengan jaringan VLAN 14, VLAN 15, VLAN 16, dan VLAN 17. R-DIST juga digunakan sebagai salah satu endpoint konfigurasi OSPF Virtual Link dengan R-Core-3.
R-ISP-1	Berfungsi sebagai router yang menghubungkan jaringan internal PT XYZ dengan jaringan luar/internet. Router ini menangani jalur menuju jaringan eksternal serta proses NAT/PAT agar perangkat pada jaringan internal dapat mengakses internet.

SW-IT-1	Berfungsi sebagai switch yang menghubungkan PC1 dan PC2 pada VLAN 11 dengan R-Core-2. Port menuju PC dikonfigurasi sebagai access port, sedangkan port menuju router digunakan sebagai trunk.
SW-IT-2	Berfungsi sebagai switch yang menghubungkan PC3 pada VLAN 12 dengan R-Core-2. Switch meneruskan frame VLAN 12 antara end device dan router melalui link trunk.
SW-HRD	Berfungsi sebagai switch yang menghubungkan PC4 dan PC5 pada VLAN 12 dengan R-Core-2. Switch meneruskan frame VLAN 12 antara end device dan router melalui link tersebut.
SW-MARKETING	Berfungsi sebagai switch yang menghubungkan PC6 pada VLAN 14 dan PC7 pada VLAN 15 dengan R-DIST. Switch memisahkan kedua jaringan menggunakan vlan dan meneruskannya ke router melalui link tersebut.
SW-FINANCE	Berfungsi sebagai switch yang menghubungkan PC8 pada VLAN 16 dan PC9 pada VLAN 17 dengan R-DIST. Switch melakukan pemisahan lalu lintas berdasarkan VLAN sebelum diteruskan menuju router.
PC1 dan PC2	Berfungsi sebagai end device pada VLAN 11 yang digunakan untuk melakukan pengujian komunikasi dalam VLAN, antar-VLAN, antar-area OSPF, dan akses menuju internet.
PC3	Berfungsi sebagai end device pada VLAN 12 yang digunakan sebagai salah satu host pengujian konektivitas antar-area OSPF.
PC4 dan PC5	Berfungsi sebagai end device pada VLAN 13 yang digunakan untuk mengirim dan menerima data serta menguji komunikasi menuju VLAN lainnya.
PC6	Berfungsi sebagai end device pada VLAN 14 dan digunakan untuk menguji konektivitas menuju VLAN lain maupun jaringan eksternal.
PC7	Berfungsi sebagai end device pada VLAN 15 dan digunakan sebagai host pada jaringan Marketing.
PC8	Berfungsi sebagai end device pada VLAN 16 dan digunakan sebagai host pada jaringan Finance.
PC9	Berfungsi sebagai end device pada VLAN 17 dan digunakan sebagai host pada jaringan Finance.
Net	Merepresentasikan jaringan luar atau internet yang digunakan sebagai tujuan pengujian konektifitas eksternal, misalnya melalui pengujian ping menuju alamat 8.8.8.8.

1. PC

Konfigurasi: Setiap PC dikonfigurasi menggunakan IP address, subnet mask, dan default gateway sesuai dengan jaringan VLAN masing-masing.

Verifikasi : #sh ip

2. Konfigurasi Dasar Perangkat

enable

configure terminal

hostname R-Core-1

```
enable secret Admin
username Admin privilege 15 secret Admin
ip domain-name telmat.net
crypto key generate rsa modulus 1024
ip ssh version 2
line vty 0 4
login local
transport input ssh
exit
end
write memory
```

3. Konfigurasi Router R-Core-1

```
enable
configure terminal
hostname R-Core-1

interface loopback0
 ip address 192.168.0.1 255.255.255.255
 ip ospf 123 area 0
 no shutdown
exit

interface e0/0
 ip address 172.16.0.2 255.255.255.252
 no shutdown
exit

interface e0/1
 ip address 192.168.112.1 255.255.255.252
 ip ospf 123 area 0
```

```
 no shutdown
exit

interface e0/2
 ip address 192.168.113.1 255.255.255.252
 ip ospf 123 area 0
 no shutdown
exit

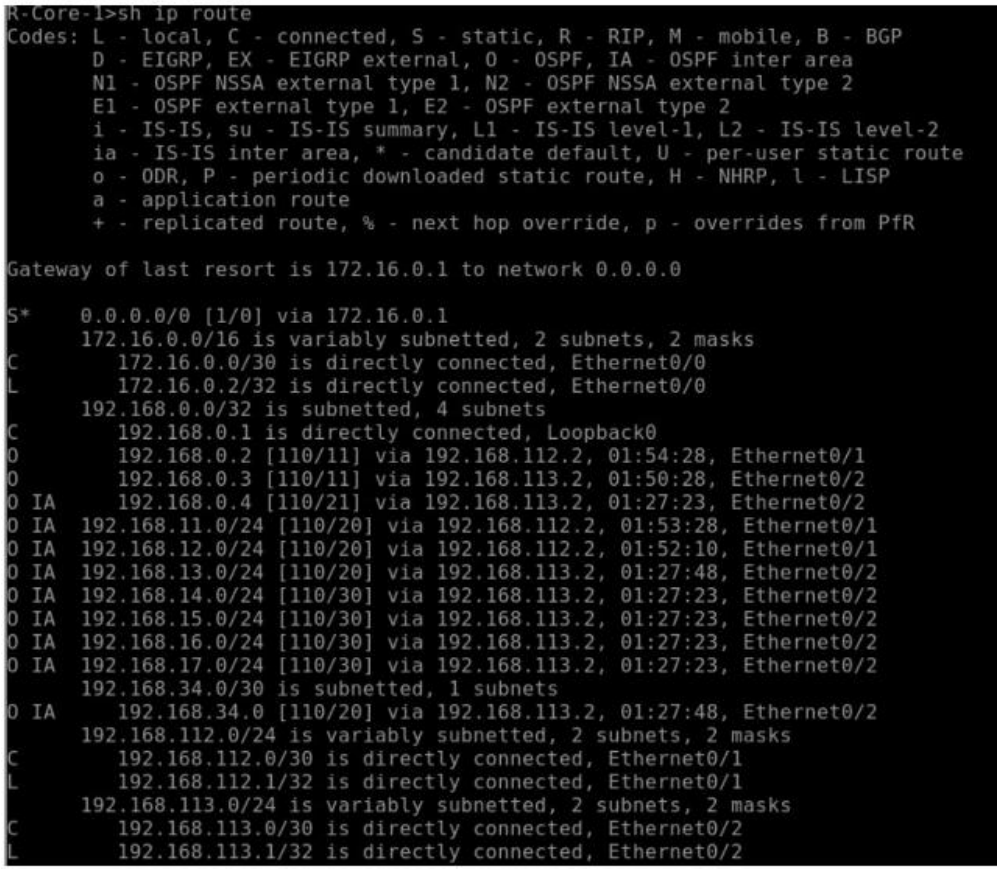
router ospf 123
 router-id 192.168.0.1
 passive-interface loopback0
exit

ip route 0.0.0.0 0.0.0.0 172.16.0.1

end
write memory
```

Verifikasi:

```
R-Core-1#show ip route
```



4) Konfigurasi R-Core-2

enable

configure terminal

hostname R-Core-2

interface loopback0

ip address 192.168.0.2 255.255.255.255

ip ospf 123 area 0

exit

interface e0/0

ip address 192.168.112.2 255.255.255.252

ip ospf 123 area 0

no shutdown

exit

```
interface e0/1
 no ip address
 no shutdown
exit

interface e0/1.11
 encapsulation dot1Q 11
 ip address 192.168.11.1 255.255.255.0
 ip ospf 123 area 1
exit

interface e0/2
 no ip address
 no shutdown
exit

interface e0/2.12
 encapsulation dot1Q 12
 ip address 192.168.12.1 255.255.255.0
 ip ospf 123 area 1
exit

router ospf 123
 router-id 192.168.0.2
 passive-interface loopback0
 passive-interface e0/1.11
 passive-interface e0/2.12
exit

end
```

write memory

Verifikasi :

```
R-Core-2>sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is 192.168.112.1 to network 0.0.0.0

O*E2  0.0.0.0/0 [110/1] via 192.168.112.1, 01:03:29, Ethernet0/0
      192.168.0.0/32 is subnetted, 4 subnets
O        192.168.0.1 [110/11] via 192.168.112.1, 01:52:54, Ethernet0/0
C        192.168.0.2 is directly connected, Loopback0
O        192.168.0.3 [110/21] via 192.168.112.1, 01:49:48, Ethernet0/0
O IA     192.168.0.4 [110/31] via 192.168.112.1, 01:26:43, Ethernet0/0
      192.168.11.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.11.0/24 is directly connected, Ethernet0/1.11
L        192.168.11.1/32 is directly connected, Ethernet0/1.11
      192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.12.0/24 is directly connected, Ethernet0/2.12
L        192.168.12.1/32 is directly connected, Ethernet0/2.12
O IA  192.168.13.0/24 [110/30] via 192.168.112.1, 01:27:08, Ethernet0/0
O IA  192.168.14.0/24 [110/40] via 192.168.112.1, 01:26:43, Ethernet0/0
O IA  192.168.15.0/24 [110/40] via 192.168.112.1, 01:26:43, Ethernet0/0
O IA  192.168.16.0/24 [110/40] via 192.168.112.1, 01:26:43, Ethernet0/0
O IA  192.168.17.0/24 [110/40] via 192.168.112.1, 01:26:43, Ethernet0/0
      192.168.34.0/30 is subnetted, 1 subnets
O IA     192.168.34.0 [110/30] via 192.168.112.1, 01:27:08, Ethernet0/0
      192.168.112.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.112.0/30 is directly connected, Ethernet0/0
L        192.168.112.2/32 is directly connected, Ethernet0/0
      192.168.113.0/30 is subnetted, 1 subnets
O        192.168.113.0 [110/20] via 192.168.112.1, 01:52:54, Ethernet0/0
```

5) Konfigurasi R-Core-3

enable

configure terminal

hostname R-Core-3

interface loopback0

 ip address 192.168.0.3 255.255.255.255

 ip ospf 123 area 0

exit

interface e0/0

 ip address 192.168.113.2 255.255.255.252

 ip ospf 123 area 0

```
 no shutdown
exit

interface e0/1
 no ip address
 no shutdown
exit

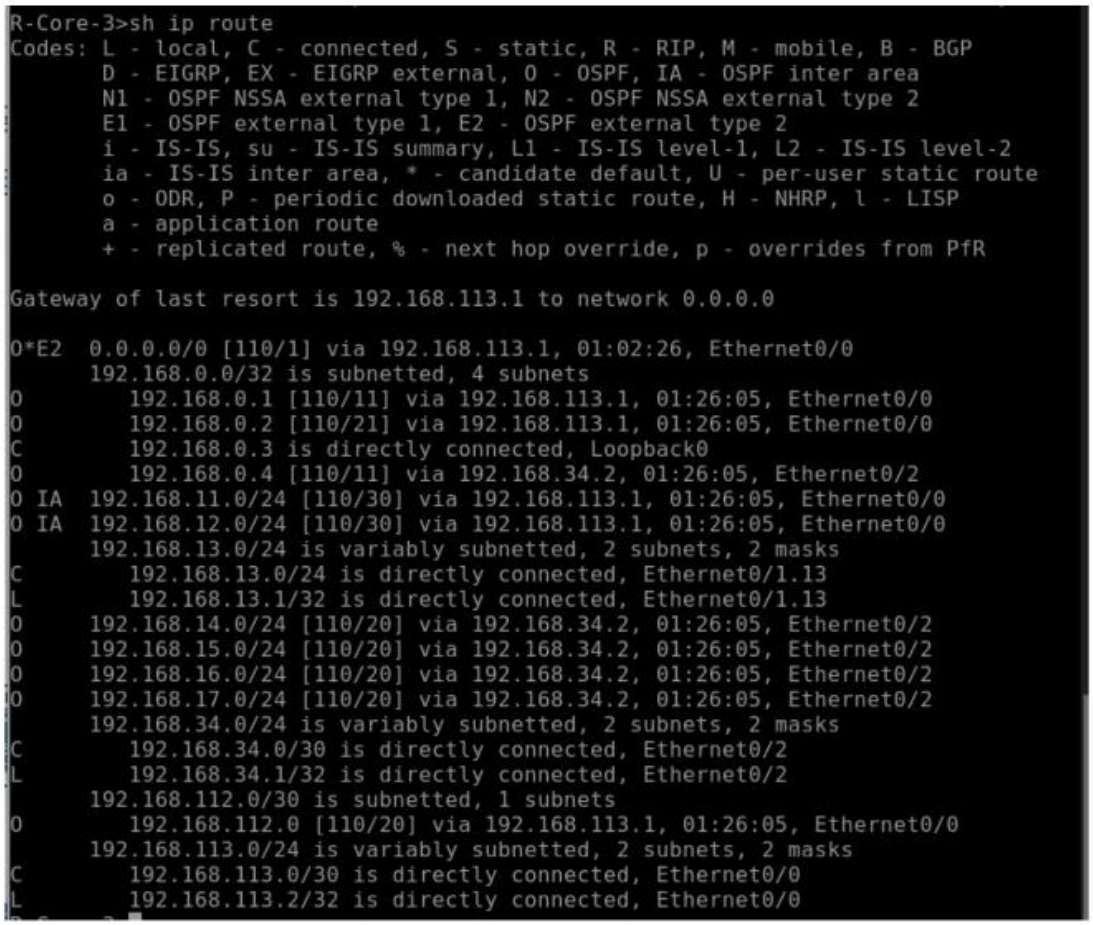
interface e0/1.13
 encapsulation dot1Q 13
 ip address 192.168.13.1 255.255.255.0
 ip ospf 123 area 1
exit

interface e0/2
 ip address 192.168.34.1 255.255.255.252
 ip ospf 123 area 2
 no shutdown
exit

router ospf 123
 router-id 192.168.0.3
 passive-interface loopback0
 passive-interface e0/1.13
exit

end
write memory
```

Verifikasi :



5. Konfigurasi R-DIST

enable

configure terminal

hostname R-DIST

interface loopback0

ip address 192.168.0.4 255.255.255.255

ip ospf 123 area 2

exit

interface e0/0

ip address 192.168.34.2 255.255.255.252

ip ospf 123 area 2

no shutdown

exit

```
interface e0/1
 no ip address
 no shutdown
exit

interface e0/1.14
 encapsulation dot1Q 14
 ip address 192.168.14.1 255.255.255.0
 ip ospf 123 area 2
exit

interface e0/1.15
 encapsulation dot1Q 15
 ip address 192.168.15.1 255.255.255.0
 ip ospf 123 area 2
exit

interface e0/2
 no ip address
 no shutdown
exit

interface e0/2.16
 encapsulation dot1Q 16
 ip address 192.168.16.1 255.255.255.0
 ip ospf 123 area 2
exit

interface e0/2.17
```

encapsulation dot1Q 17

ip address 192.168.17.1 255.255.255.0

ip ospf 123 area 2

exit

router ospf 123

router-id 192.168.0.4

passive-interface loopback0

passive-interface e0/1.14

passive-interface e0/1.15

passive-interface e0/2.16

passive-interface e0/2.17

exit

end

write memory

```
R-DIST>sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is 192.168.34.1 to network 0.0.0.0

O*E2  0.0.0.0/0 [110/1] via 192.168.34.1, 01:01:48, Ethernet0/0
      192.168.0.0/32 is subnetted, 4 subnets
O        192.168.0.1 [110/21] via 192.168.34.1, 01:25:07, Ethernet0/0
O        192.168.0.2 [110/31] via 192.168.34.1, 01:25:07, Ethernet0/0
O        192.168.0.3 [110/11] via 192.168.34.1, 01:25:07, Ethernet0/0
C        192.168.0.4 is directly connected, Loopback0
O IA  192.168.11.0/24 [110/40] via 192.168.34.1, 01:25:07, Ethernet0/0
O IA  192.168.12.0/24 [110/40] via 192.168.34.1, 01:25:07, Ethernet0/0
O IA  192.168.13.0/24 [110/20] via 192.168.34.1, 01:25:07, Ethernet0/0
      192.168.14.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.14.0/24 is directly connected, Ethernet0/1.14
L        192.168.14.1/32 is directly connected, Ethernet0/1.14
      192.168.15.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.15.0/24 is directly connected, Ethernet0/1.15
L        192.168.15.1/32 is directly connected, Ethernet0/1.15
      192.168.16.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.16.0/24 is directly connected, Ethernet0/2.16
L        192.168.16.1/32 is directly connected, Ethernet0/2.16
      192.168.17.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.17.0/24 is directly connected, Ethernet0/2.17
L        192.168.17.1/32 is directly connected, Ethernet0/2.17
      192.168.34.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.34.0/30 is directly connected, Ethernet0/0
L        192.168.34.2/32 is directly connected, Ethernet0/0
      192.168.112.0/30 is subnetted, 1 subnets
O        192.168.112.0 [110/30] via 192.168.34.1, 01:25:07, Ethernet0/0
      192.168.113.0/30 is subnetted, 1 subnets
O        192.168.113.0 [110/20] via 192.168.34.1, 01:25:07, Ethernet0/0
```

6) Konfigurasi SW-IT-1

```
enable
configure terminal
hostname SW-IT-1

vlan 11
 name IT
exit

interface e0/0
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 11
 no shutdown
exit

interface e1/0
 switchport mode access
 switchport access vlan 11
 spanning-tree portfast
 no shutdown
exit

interface e1/2
 switchport mode access
 switchport access vlan 11
 spanning-tree portfast
 no shutdown
exit
```

```
end
write memory
```

7. Konfigurasi SW-IT-2

```
enable
configure terminal
hostname SW-IT-2

vlan 12
 name IT2
exit

interface e0/0
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 12
 no shutdown
exit

interface e1/0
 switchport mode access
 switchport access vlan 12
 spanning-tree portfast
 no shutdown
exit

end
write memory
```

8. Konfigurasi SW-HRD

```
enable
configure terminal
hostname SW-HRD

vlan 13
 name HRD
exit

interface e0/0
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 13
 no shutdown
exit

interface e1/0
 switchport mode access
 switchport access vlan 13
 spanning-tree portfast
 no shutdown
exit

interface e1/2
 switchport mode access
 switchport access vlan 13
 spanning-tree portfast
 no shutdown
exit
```

```
end
write memory

9. Konfigurasi SW-Marketing
enable
configure terminal
hostname SW-MARKETING

vlan 14
 name MARKETING
vlan 15
 name MARKETING2
exit

interface e0/0
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 14,15
 no shutdown
exit

interface e1/0
 switchport mode access
 switchport access vlan 14
 spanning-tree portfast
 no shutdown
exit

interface e2/0
 switchport mode access
```

```
 switchport access vlan 15
 spanning-tree portfast
 no shutdown
exit

end
write memory
```

10. Konfigurasi SW-Finance

```
enable
configure terminal
hostname SW-FINANCE

vlan 16
 name FINANCE
vlan 17
 name FINANCE2
exit

interface e0/0
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 16,17
 no shutdown
exit

interface e1/0
 switchport mode access
 switchport access vlan 16
 spanning-tree portfast
```

```
 no shutdown
exit

interface e2/0
 switchport mode access
 switchport access vlan 17
 spanning-tree portfast
 no shutdown
exit

end
write memory
```

11. Pada R-Core-3

```
configure terminal
router ospf 123
 area 2 virtual-link 192.168.0.4
end
write memory
```

12. Pada R-DIST

```
configure terminal
router ospf 123
 area 2 virtual-link 192.168.0.3
end
write memory
```

13. Pada R-ISP-1

```
enable
configure terminal
hostname R-ISP-1

interface loopback0
 ip address 1.1.1.1 255.255.255.255
exit

interface e0/1
 ip address 172.16.0.1 255.255.255.252
 no shutdown
exit

interface e0/0
 ip address dhcp
 ip nat outside
 no shutdown
exit

ip route 192.168.0.0 255.255.0.0 172.16.0.2

access-list 1 permit 192.168.0.0 0.0.255.255
access-list 1 permit 192.168.11.0 0.0.0.255
access-list 1 permit 192.168.12.0 0.0.0.255
access-list 1 permit 192.168.13.0 0.0.0.255
access-list 1 permit 192.168.14.0 0.0.0.255
access-list 1 permit 192.168.15.0 0.0.0.255
access-list 1 permit 192.168.16.0 0.0.0.255
access-list 1 permit 192.168.17.0 0.0.0.255
```

```
interface e0/1
 ip nat inside
exit

ip nat inside source list 1 interface e0/0 overload

end
write memory
```